Sonu Kumar

Bachelor of Technology
Electrical Engineering
National Institute of Technology, Patna

+91-6299710356
sonuk.ug23.ee@nitp.ac.in
linkedin.com/in/Sonu Kumar
github.com/sumantpatel37819

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., EE	National Institute of Technology, Patna	7.64	2023-2027
Senior Secondary	BSEB	91.2	2022
Secondary	CBSE	88.6	2020

Experience

- **Research Project – AI-Powered Transmission Line Inspection Robot** *NIT Patna*
  Computer Vision & AI for Power Systems Dec 2024 – June 2025
  - Built scalable pipelines for **image-based anomaly detection**, applying ML models to identify cracks, wear, and foreign objects in large datasets.
  - Implemented **feature extraction and embeddings** from sensor & vision data, enabling predictive maintenance and prioritization of inspections.
- **House Price Prediction Web App**
  Flask · Python · HTML/CSS/JS · Machine Learning GitHub Repo
  - Created a Flask-based web app for real-time house price prediction using a trained ML model.
  - Designed RESTful endpoint, responsive UI, and integrated error handling sample data loading for enhanced UX.

Projects

- **TourKaro – AI-Powered Tourist Guide Web Application** *Jul 2025 – Present*
  Flask, Python, HTML, CSS, JavaScript, Google Gemini API, REST API Project Link
  - Developed an **AI-driven travel assistant** using Flask and Gemini API to provide **real-time tourist guidance and recommendations**.
  - Built a RESTful backend and responsive frontend enabling personalized trip planning and location-based insights.
  - Integrated an AI chat interface for interactive and intelligent travel assistance, enhancing user engagement.
- **Mini Projects**
  OpenCV, Hugging Face, PyTorch, Scikit-learn
  - Implemented a **Face Detection System** in Python (OpenCV) for real-time recognition.
  - Developed a persona-based chatbot using Hugging Face & DialoGPT with PyTorch, improving conversational relevance.

Skills

- **Programming**: Python, C, C++, Java, SQL
- **ML/DL Frameworks**: TensorFlow, PyTorch, scikit-learn, Keras
- **Data Science Tools**: Pandas, NumPy, Matplotlib, Seaborn
- **Core ML Concepts**: Probability & Statistics, Linear Algebra, Calculus, NLP, Computer Vision
- **Soft Skills**: Leadership, Teamwork, Communication, Project Management

Certifications

- **NIT Patna**, Laser Bot Race Challenge *Jan 2024*
- **NIT Patna**, Tinkering Lab Event *Jan 2024*

Achievements

- Led a team in IRoC-U Challenge 2025, developing a functional drone that advanced through two rounds.
- Ranked 3rd in district, Intermediate Science (2022)
- Won Gold in Laser Bot Race, Silver in Tinkering Lab, and Bronze in Grand Line Challenge

Positions of Responsibility

- **Vice President,**Robotics Club, NIT Patna *2025 – Present*
- **Technical Team Member,** Robotics Club, NIT Patna *2023 – 2025*